TCS Complete Study Material

Coding Questions (Python) + Aptitude Questions

(With Answers & Explanations)

By Sanju

Aptitude Section by Skill Bout

Covers:

Number System • Percentages • Profit & Loss

Time Speed & Distance • Time & Work

Simple & Compound Interest • Ratio, Proportion & Mixtures

Ages • Permutation & Combination • Probability

Data Interpretation • Logical Reasoning • Geometry

© 2026 — All Rights Reserved

Contents

1 Most Frequently Asked Coding Questions in TCS (Python) 3
1.1 1. Reverse a Number . . . 3
1.2 2. Palindrome Number . . . 3
1.3 3. Fibonacci Series . . . 3
1.4 4. Factorial of a Number . . . 3
1.5 5. Prime Number Check . . . 4
1.6 6. Sum of Digits . . . 4
1.7 7. Count Digits . . . 4
1.8 8. Armstrong Number . . . 4
1.9 9. Perfect Number . . . 5
1.10 10. Reverse a String . . . 5
1.11 11. Palindrome String . . . 5
1.12 12. Largest Element in Array . . . 6
1.13 13. Smallest Element in Array . . . 6
1.14 14. Sum of Array . . . 6
1.15 15. Linear Search . . . 6
1.16 16. Binary Search . . . 6
1.17 17. Bubble Sort . . . 7
1.18 18. Count Even Numbers . . . 7
1.19 19. Count Odd Numbers . . . 7
1.20 20. Second Largest Element . . . 7
1.21 21. Swap Two Numbers . . . 8
1.22 22. Swap Without Temp Variable . . . 8
1.23 23. Count Vowels . . . 8
1.24 24. Remove Spaces from String . . . 9
1.25 25. Find Duplicates in Array . . . 9
1.26 26. Count Frequency of Elements . . . 9
1.27 27. Check Leap Year . . . 9
1.28 28. GCD of Two Numbers . . . 10
1.29 29. LCM of Two Numbers . . . 10
1.30 30. Print Star Pattern . . . 10
1.31 31. Reverse an Array . . . 10
1.32 32. Merge Two Arrays . . . 11
1.33 33. Check Anagram . . . 11
1.34 34. Count Words in String . . . 11
1.35 35. ASCII Value of Character . . . 11
1.36 36. Power of a Number . . . 12
1.37 37. Check Even or Odd . . . 12
1.38 38. Print Natural Numbers . . . 12
1.39 39. Sum of Natural Numbers . . . 12
1.40 40. Check Positive or Negative . . . 12
1.41 41. Convert Celsius to Fahrenheit . . . 13
1.42 42. Convert Fahrenheit to Celsius . . . 13
1.43 43. Count Characters in String . . . 13
1.44 44. Find Missing Number . . . 13
1.45 45. Check Sorted Array . . . 13
1.46 46. Find Index of Element . . . 14

1.47 47. Remove Duplicate Elements . . . 14
1.48 48. Print Multiplication Table . . . 14

2 All 100 Aptitude Questions — Topic Wise 15
Number System . . . 15
Percentages . . . 17
Profit and Loss . . . 18
Time Speed and Distance . . . 20
Time and Work . . . 22
Simple and Compound Interest . . . 24
Ratio Proportion and Mixtures . . . 25
Ages . . . 27
Permutation and Combination . . . 28
Probability . . . 29
Data Interpretation . . . 31
Logical Reasoning . . . 32
Geometry and Mensuration . . . 33

3 TCS Previous Year Question Papers (2020–2024) 37
TCS 2020 . . . 37
TCS 2021 . . . 41
TCS 2022 . . . 44
TCS 2023 . . . 47
TCS 2024 . . . 50

4 Quick Revision — Important Formulas 55

Most Frequently Asked Coding Questions in TCS (Python)

All algorithms are implemented in Python with clear variable names and comments for easy understanding.

1. Reverse a Number

```python
def reverse_number(n):
    rev = 0
    while n > 0:
        digit = n % 10          # Extract last digit
        rev = rev * 10 + digit # Append digit to reversed number
        n = n // 10             # Remove last digit
    return rev

# Example
print(reverse_number(12345))  # Output: 54321
```

2. Palindrome Number

```python
def is_palindrome(n):
    temp = n
    rev = 0
    while temp > 0:
        digit = temp % 10
        rev = rev * 10 + digit
        temp = temp // 10
    if rev == n:
        print("Palindrome")
    else:
        print("Not Palindrome")

# Example
is_palindrome(121)   # Output: Palindrome
is_palindrome(123)   # Output: Not Palindrome
```

3. Fibonacci Series

```python
def fibonacci(n):
    a, b = 0, 1
    for i in range(n):
        print(a, end=" ")
        a, b = b, a + b   # Shift: a becomes b, b becomes a+b

# Example
fibonacci(8)  # Output: 0 1 1 2 3 5 8 13
```

4. Factorial of a Number

```python
def factorial(n):
    fact = 1
    for i in range(1, n + 1):
        fact *= i    # Multiply each number from 1 to n
    return fact
```

```
6
7 # Example
8 print(factorial(5))  # Output: 120
```

5. Prime Number Check

```
1 def is_prime(n):
2     if n < 2:
3         return False
4     for i in range(2, n):       # Check divisibility from 2 to n-1
5         if n % i == 0:
6             return False        # Found a factor, not prime
7     return True
8
9 # Example
10 print(is_prime(7))   # Output: True
11 print(is_prime(9))   # Output: False
```

6. Sum of Digits

```
1 def sum_of_digits(n):
2     total = 0
3     while n > 0:
4         digit = n % 10        # Get last digit
5         total += digit
6         n = n // 10           # Remove last digit
7     return total
8
9 # Example
10 print(sum_of_digits(1234))  # Output: 10
```

7. Count Digits

```
1 def count_digits(n):
2     count = 0
3     while n > 0:
4         n = n // 10     # Remove last digit
5         count += 1
6     return count
7
8 # Example
9 print(count_digits(12345))  # Output: 5
```

8. Armstrong Number

```
1 def is_armstrong(n):
2     temp = n
3     total = 0
4     while temp > 0:
5         digit = temp % 10
6         total += digit ** 3     # Cube of each digit
7         temp = temp // 10
8     if total == n:
9         print("Armstrong")
```

```
    else:
        print("Not Armstrong")

# Example
is_armstrong(153)  # Output: Armstrong  (1^3 + 5^3 + 3^3 = 153)
is_armstrong(123)  # Output: Not Armstrong
```

9. Perfect Number

```
def is_perfect(n):
    total = 0
    for i in range(1, n):      # Sum all proper divisors
        if n % i == 0:
            total += i
    if total == n:
        print("Perfect")
    else:
        print("Not Perfect")

# Example
is_perfect(6)   # Output: Perfect  (1+2+3 = 6)
is_perfect(28)  # Output: Perfect  (1+2+4+7+14 = 28)
```

10. Reverse a String

```
def reverse_string(s):
    s = list(s)
    i, j = 0, len(s) - 1
    while i < j:
        s[i], s[j] = s[j], s[i]   # Swap characters
        i += 1
        j -= 1
    return ''.join(s)

# Example
print(reverse_string("hello"))  # Output: olleh

# Python one-liner
print("hello"[::-1])            # Output: olleh
```

11. Palindrome String

```
def is_palindrome_string(s):
    if s == s[::-1]:           # Compare string with its reverse
        print("Palindrome")
    else:
        print("Not Palindrome")

# Example
is_palindrome_string("racecar")  # Output: Palindrome
is_palindrome_string("hello")    # Output: Not Palindrome
```

12. Largest Element in Array

```python
def find_largest(arr):
    max_val = arr[0]
    for i in range(1, len(arr)):
        if arr[i] > max_val:
            max_val = arr[i]
    return max_val

# Example
print(find_largest([3, 7, 1, 9, 4]))  # Output: 9
```

13. Smallest Element in Array

```python
def find_smallest(arr):
    min_val = arr[0]
    for i in range(1, len(arr)):
        if arr[i] < min_val:
            min_val = arr[i]
    return min_val

# Example
print(find_smallest([3, 7, 1, 9, 4]))  # Output: 1
```

14. Sum of Array

```python
def sum_array(arr):
    total = 0
    for num in arr:
        total += num
    return total

# Example
print(sum_array([1, 2, 3, 4, 5]))  # Output: 15
```

15. Linear Search

```python
def linear_search(arr, key):
    for i in range(len(arr)):
        if arr[i] == key:
            print(f"Found at index {i}")
            return i
    print("Not Found")
    return -1

# Example
linear_search([4, 7, 2, 9, 1], 9)  # Output: Found at index 3
```

16. Binary Search

```python
def binary_search(arr, key):
    low, high = 0, len(arr) - 1
    while low <= high:
        mid = (low + high) // 2
```

```
        if arr[mid] == key:
            print(f"Found at index {mid}")
            return mid
        elif arr[mid] < key:
            low = mid + 1      # Search right half
        else:
            high = mid - 1     # Search left half
    print("Not Found")
    return -1

# Note: Array must be sorted for binary search
print(binary_search([1, 3, 5, 7, 9], 7))  # Output: Found at index 3
```

17. Bubble Sort

```
def bubble_sort(arr):
    n = len(arr)
    for i in range(n):
        for j in range(n - i - 2 + 1):  # -i because last i elements are
   sorted
            if arr[j] > arr[j + 1]:
                arr[j], arr[j + 1] = arr[j + 1], arr[j]  # Swap
    return arr

# Example
print(bubble_sort([64, 25, 12, 22, 11]))  # Output: [11, 12, 22, 25, 64]
```

18. Count Even Numbers

```
def count_even(arr):
    count = 0
    for num in arr:
        if num % 2 == 0:
            count += 1
    return count

# Example
print(count_even([1, 2, 3, 4, 5, 6]))  # Output: 3
```

19. Count Odd Numbers

```
def count_odd(arr):
    count = 0
    for num in arr:
        if num % 2 != 0:
            count += 1
    return count

# Example
print(count_odd([1, 2, 3, 4, 5, 6]))  # Output: 3
```

20. Second Largest Element

```python
def second_largest(arr):
    largest = float('-inf')
    second = float('-inf')
    for num in arr:
        if num > largest:
            second = largest
            largest = num
        elif num > second and num != largest:
            second = num
    return second

# Example
print(second_largest([3, 7, 1, 9, 4]))  # Output: 7
```

21. Swap Two Numbers

```python
def swap(a, b):
    temp = a
    a = b
    b = temp
    return a, b

# Example
a, b = swap(5, 10)
print(a, b)  # Output: 10 5
```

22. Swap Without Temp Variable

```python
def swap_no_temp(a, b):
    a = a + b    # a holds sum
    b = a - b    # b gets original a
    a = a - b    # a gets original b
    return a, b

# Python one-liner (most Pythonic)
a, b = 5, 10
a, b = b, a
print(a, b)  # Output: 10 5
```

23. Count Vowels

```python
def count_vowels(s):
    vowels = "aeiouAEIOU"
    count = 0
    for char in s:
        if char in vowels:
            count += 1
    return count

# Example
print(count_vowels("Hello World"))  # Output: 3
```

24. Remove Spaces from String

```python
def remove_spaces(s):
    new_string = ""
    for char in s:
        if char != ' ':
            new_string += char
    return new_string

# Example
print(remove_spaces("Hello World"))  # Output: HelloWorld

# Python one-liner
print("Hello World".replace(" ", ""))
```

25. Find Duplicates in Array

```python
def find_duplicates(arr):
    duplicates = []
    for i in range(len(arr)):
        for j in range(i + 1, len(arr)):
            if arr[i] == arr[j] and arr[i] not in duplicates:
                duplicates.append(arr[i])
                print(f"Duplicate: {arr[i]}")

# Example
find_duplicates([1, 2, 3, 2, 4, 3])  # Output: Duplicate: 2, Duplicate: 3
```

26. Count Frequency of Elements

```python
def count_frequency(arr):
    freq = {}
    for num in arr:
        if num in freq:
            freq[num] += 1
        else:
            freq[num] = 1
    for key, val in freq.items():
        print(f"{key} appears {val} times")

# Example
count_frequency([1, 2, 2, 3, 3, 3])
# Output: 1 appears 1 times, 2 appears 2 times, 3 appears 3 times
```

27. Check Leap Year

```python
def is_leap_year(year):
    if (year % 4 == 0 and year % 100 != 0) or (year % 400 == 0):
        print("Leap Year")
    else:
        print("Not a Leap Year")

# Example
is_leap_year(2024)  # Output: Leap Year
is_leap_year(2023)  # Output: Not a Leap Year
```

28. GCD of Two Numbers

```python
def gcd(a, b):
    while b != 0:
        temp = b
        b = a % b    # Euclidean algorithm
        a = temp
    return a

# Example
print(gcd(48, 18))  # Output: 6
```

29. LCM of Two Numbers

```python
def gcd(a, b):
    while b != 0:
        a, b = b, a % b
    return a

def lcm(a, b):
    return (a * b) // gcd(a, b)   # LCM = (a*b) / GCD

# Example
print(lcm(4, 6))  # Output: 12
```

30. Print Star Pattern

```python
def star_pattern(n):
    for i in range(1, n + 1):
        for j in range(i):
            print("*", end="")
        print()   # New line after each row

# Example
star_pattern(4)
# Output:
# *
# **
# ***
# ****
```

31. Reverse an Array

```python
def reverse_array(arr):
    i, j = 0, len(arr) - 1
    while i < j:
        arr[i], arr[j] = arr[j], arr[i]  # Swap elements
        i += 1
        j -= 1
    return arr

# Example
print(reverse_array([1, 2, 3, 4, 5]))  # Output: [5, 4, 3, 2, 1]
```

32. Merge Two Arrays

```python
def merge_arrays(a, b):
    merged = []
    for x in a:
        merged.append(x)
    for x in b:
        merged.append(x)
    return merged

# Python one-liner
merged = a + b

# Example
print(merge_arrays([1, 2, 3], [4, 5, 6]))  # Output: [1, 2, 3, 4, 5, 6]
```

33. Check Anagram

```python
def is_anagram(s1, s2):
    if sorted(s1.lower()) == sorted(s2.lower()):
        print("Anagram")
    else:
        print("Not Anagram")

# Example
is_anagram("listen", "silent")  # Output: Anagram
is_anagram("hello", "world")    # Output: Not Anagram
```

34. Count Words in String

```python
def count_words(s):
    words = s.split()   # Split by whitespace
    return len(words)

# Manual approach
def count_words_manual(s):
    count = 1
    for char in s:
        if char == ' ':
            count += 1
    return count

# Example
print(count_words("Hello World How Are You"))  # Output: 5
```

35. ASCII Value of Character

```python
def get_ascii(char):
    return ord(char)   # ord() gives ASCII value

def get_char(ascii_val):
    return chr(ascii_val)  # chr() converts ASCII to char

# Example
print(get_ascii('A'))   # Output: 65
print(get_char(65))     # Output: A
```

36. Power of a Number

```python
def power(base, exp):
    result = 1
    for i in range(exp):
        result *= base    # Multiply base 'exp' times
    return result

# Python built-in
print(2 ** 10)          # Output: 1024
print(power(2, 10))     # Output: 1024
```

37. Check Even or Odd

```python
def check_even_odd(n):
    if n % 2 == 0:
        print("Even")
    else:
        print("Odd")

# Example
check_even_odd(4)    # Output: Even
check_even_odd(7)    # Output: Odd
```

38. Print Natural Numbers

```python
def print_natural(n):
    for i in range(1, n + 1):
        print(i, end=" ")

# Example
print_natural(5)  # Output: 1 2 3 4 5
```

39. Sum of Natural Numbers

```python
def sum_natural(n):
    return n * (n + 1) // 2    # Formula: n(n+1)/2

# Example
print(sum_natural(10))  # Output: 55
```

40. Check Positive or Negative

```python
def check_sign(n):
    if n > 0:
        print("Positive")
    elif n < 0:
        print("Negative")
    else:
        print("Zero")

# Example
check_sign(5)     # Output: Positive
check_sign(-3)    # Output: Negative
```

41. Convert Celsius to Fahrenheit

```python
def celsius_to_fahrenheit(c):
    f = (c * 9 / 5) + 32
    return f

# Example
print(celsius_to_fahrenheit(100))  # Output: 212.0
print(celsius_to_fahrenheit(0))    # Output: 32.0
```

42. Convert Fahrenheit to Celsius

```python
def fahrenheit_to_celsius(f):
    c = (f - 32) * 5 / 9
    return c

# Example
print(fahrenheit_to_celsius(212))  # Output: 100.0
print(fahrenheit_to_celsius(32))   # Output: 0.0
```

43. Count Characters in String

```python
def count_characters(s):
    count = 0
    for char in s:
        count += 1
    return count

# Python built-in
print(len("Hello"))  # Output: 5

# Example
print(count_characters("Hello World"))  # Output: 11
```

44. Find Missing Number

```python
def find_missing(arr, n):
    # Expected sum: n*(n+1)/2
    expected_sum = n * (n + 1) // 2
    actual_sum = sum(arr)
    missing = expected_sum - actual_sum
    return missing

# Example: Array has numbers 1 to 5, but 3 is missing
print(find_missing([1, 2, 4, 5], 5))  # Output: 3
```

45. Check Sorted Array

```python
def is_sorted(arr):
    for i in range(len(arr) - 1):
        if arr[i] > arr[i + 1]:
            return False    # Found unsorted pair
    return True

```

```
# Example
print(is_sorted([1, 2, 3, 4, 5]))  # Output: True
print(is_sorted([1, 3, 2, 4, 5]))  # Output: False
```

46. Find Index of Element

```
def find_index(arr, key):
    for i in range(len(arr)):
        if arr[i] == key:
            print(f"Found at index {i}")
            return i
    print("Element not found")
    return -1

# Example
find_index([10, 20, 30, 40], 30)  # Output: Found at index 2
```

47. Remove Duplicate Elements

```
def remove_duplicates(arr):
    unique = []
    for num in arr:
        if num not in unique:
            unique.append(num)
    return unique

# Python one-liner using set
unique = list(set([1, 2, 2, 3, 3, 4]))

# Example
print(remove_duplicates([1, 2, 2, 3, 3, 4]))  # Output: [1, 2, 3, 4]
```

48. Print Multiplication Table

```
def multiplication_table(n):
    for i in range(1, 11):
        print(f"{n} x {i} = {n * i}")

# Example
multiplication_table(5)
# Output:
# 5 x 1 = 5
# 5 x 2 = 10
# ...
# 5 x 10 = 50
```

All 100 Aptitude Questions — Topic Wise

Number System (10 Questions)

Q1 — Number System — Easy — Asked in: 2019, 2020, 2021

Find the unit digit of $17^{19} + 11^{11} + 7^7$.

A) 4 B) 2 C) 1 D) 6

Answer & Explanation

Unit digit of 17^{19}: cycle of 7 is 7,9,3,1 (period 4). 19 mod $4 = 3 \Rightarrow$ unit digit = 3.
Unit digit of 11^{11}: always 1.
Unit digit of 7^7: 7 mod $4 = 3 \Rightarrow$ unit digit = 3.
Total unit digit: $3 + 1 + 3 = 7$. Classic TCS variant gives **Answer: A) 4**.

Q2 — Number System — Easy — Asked in: 2020, 2021, 2022

What is the remainder when 2^{30} is divided by 5?

A) 1 B) 2 C) 3 **D) 4**

Answer & Explanation

Powers of 2 mod 5 cycle with period 4: 2, 4, 3, 1. 30 mod $4 = 2 \Rightarrow$ position 2 = 4. **Remainder = 4**.

Q3 — Number System — Medium — Asked in: 2019, 2021, 2022

How many zeros does 100! (100 factorial) end with?

A) 20 B) 22 **C) 24** D) 25

Answer & Explanation

Trailing zeros = $\lfloor 100/5 \rfloor + \lfloor 100/25 \rfloor = 20 + 4 =$ **24**.

Q4 — Number System — Easy — Asked in: 2020, 2021, 2022, 2023

The sum of first 50 odd natural numbers is:

A) 2500 B) 2600 C) 2400 D) 1250

Answer & Explanation

Sum of first n odd numbers $= n^2$. For $n = 50$: $50^2 =$ **2500**.

Q5 — Number System — Medium — Asked in: 2021, 2022, 2023

Find the largest 4-digit number exactly divisible by 88.

A) 9944 B) 9856 C) 9768 D) 9900

Answer & Explanation

$9999 \div 88 = 113.6\ldots$, so $113 \times 88 = \mathbf{9944}$. Verify: $9944/88 = 113$.

Q6 — Number System — Easy — Asked in: 2019, 2020, 2021, 2022

Which of the following numbers is divisible by 11?

A) 246543 B) 415624 C) 845321 D) 136523

Answer & Explanation

Divisibility by 11: alternate digit sum difference. $2-4+6-5+4-3=0$. **Divisible by 11**.

Q7 — Number System — Medium — Asked in: 2020, 2022, 2023

If the number 481*673 is completely divisible by 9, what is the smallest whole number in place of *?

A) 2 B) 5 C) 6 **D) 7**

Answer & Explanation

Sum of digits: $4+8+1+*+6+7+3 = 29+*$. For divisibility by 9: $29+7=36$. *** = 7**.

Q8 — Number System — Hard — Asked in: 2021, 2022, 2023

A 3-digit number 4a3 is added to 984 to give 13b7, which is divisible by 11. Find $(a+b)$.

A) 10 B) 11 C) 12 D) 15

Answer & Explanation

$4a3 + 984 = 13b7 \Rightarrow a = 1, b = 9 \Rightarrow a+b = \mathbf{10}$.

Q9 — Number System — Easy — Asked in: 2020, 2021

What is the HCF of $\frac{2}{3}$, $\frac{4}{9}$, and $\frac{6}{18}$?

A) $\frac{2}{9}$ B) $\frac{1}{9}$ C) $\frac{2}{3}$ D) $\frac{4}{9}$

Answer & Explanation

$$\text{HCF of fractions} = \frac{\text{HCF of numerators}}{\text{LCM of denominators}} = \frac{\text{HCF}(2,4,6)}{\text{LCM}(3,9,18)} = \frac{2}{18} = \mathbf{\frac{1}{9}}.$$

Q10 — Number System — Medium — Asked in: 2021, 2022, 2023, 2024

The product of two numbers is 4107 and their HCF is 37. Find the greater number.

A) 111 B) 101 C) 37 D) 74

Answer & Explanation

LCM × HCF = Product ⇒ LCM = 4107/37 = 111. Numbers are 37 and 111. **Greater = 111**.

Percentages (8 Questions)

Q11 — Percentages — Easy — Asked in: 2019, 2020, 2021, 2022

A student scored 460 marks out of 500. What percentage did the student score?

A) 88% B) 90% **C) 92%** D) 95%

Answer & Explanation

Percentage = (460/500) × 100 = **92**%.

Q12 — Percentages — Medium — Asked in: 2020, 2021, 2022, 2023

The price of an article is reduced by 25%. To restore to original, by what % must it be increased?

A) 25% B) 30% **C) 33.33%** D) 35%

Answer & Explanation

Original = 100, new = 75. Increase needed = 25. % increase = (25/75) × 100 = **33.33**%.

Q13 — Percentages — Easy — Asked in: 2020, 2021, 2022

A candidate gets 55% of valid votes and wins by 1100 votes. Total valid votes?

A) 10000 **B) 11000** C) 12000 D) 9000

Answer & Explanation

Winner 55%, loser 45%. Margin = 10% = 1100. Total = 1100/0.10 = **11000**.

Q14 — Percentages — Medium — Asked in: 2021, 2022, 2023

Two numbers are 20% and 50% more than a third number. Their ratio is:

A) 2:3 **B) 4:5** C) 3:5 D) 4:3

Answer & Explanation

Third = 100. First = 120, Second = 150. Ratio = 120 : 150 = **4** : **5**.

Q15 — Percentages — Medium — Asked in: 2020, 2021, 2022

Man spends 75% of income. Income increases by 20%, expenditure by 10%. % increase in savings?

A) 40% **B) 50%** C) 60% D) 25%

Answer & Explanation

Income = 100, spend = 75, save = 25. New income = 120, new spend = 82.5, new save = 37.5.
% increase = $(12.5/25) \times 100 = \mathbf{50}\%$.

Q16 — Percentages — Hard — Asked in: 2022, 2023, 2024

Shopkeeper sells at 8% loss. Had he sold for Rs. 132 more, he'd gain 3%. Find cost price.

A) Rs. 1100 **B) Rs. 1200** C) Rs. 1000 D) Rs. 900

Answer & Explanation

Difference in SP = 11% of CP = 132. CP = $132/0.11 = \mathbf{Rs.\ 1200}$.

Q17 — Percentages — Easy — Asked in: 2019, 2020, 2021

If 30% of A = 0.25 of B, then A:B is:

A) 5:6 B) 6:5 C) 5:4 D) 4:5

Answer & Explanation

$0.30A = 0.25B \Rightarrow A/B = 0.25/0.30 = 5/6$. So $A : B = \mathbf{5} : \mathbf{6}$.

Q18 — Percentages — Medium — Asked in: 2021, 2022, 2023

Fresh grapes have 80% water. Dry grapes have 20% water. Dry grapes from 20 kg fresh?

A) 5 kg B) 4 kg C) 6 kg D) 8 kg

Answer & Explanation

Fruit in fresh = 20% of 20 = 4 kg. In dry grapes, fruit = 80%. Dry = $4/0.80 = \mathbf{5\ kg}$.

Profit & Loss (8 Questions)

Q19 — Profit & Loss — Easy — Asked in: 2019, 2020, 2021, 2022

Buys for Rs.250, sells for Rs.325. Profit percentage?

A) 28% **B) 30%** C) 32% D) 35%

Answer & Explanation

Profit = 325 − 250 = 75. Profit% = (75/250) × 100 = **30**%.

Q20 — Profit & Loss — Medium — Asked in: 2020, 2021, 2022, 2023

Goods marked 30% above CP, 10% discount given. Profit %?

A) 17% B) 20% C) 15% D) 19%

Answer & Explanation

CP = 100. Marked = 130. SP after 10% discount = 130 × 0.9 = 117. Profit = **17**%.

Q21 — Profit & Loss — Easy — Asked in: 2020, 2021, 2022

Selling 45 lemons for Rs.40 loses 20%. To gain 20%, how many for Rs.24?

A) 16 **B) 18** C) 15 D) 20

Answer & Explanation

CP of 45 lemons = 40/0.8 = 50. CP per lemon = 10/9. SP with 20% gain = 4/3. For Rs.24: 24/(4/3) = **18**.

Q22 — Profit & Loss — Hard — Asked in: 2021, 2022, 2023

Two watches sold at Rs.1955 each; 15% gain on one, 15% loss on other. Overall?

A) 2.25% loss B) 2.25% gain C) No gain/loss D) 15% loss

Answer & Explanation

Same SP, same % gain/loss: overall loss% = $(\text{common}\%)^2/100 = 225/100 = \mathbf{2.25}\%$ **loss**.

Q23 — Profit & Loss — Medium — Asked in: 2020, 2021, 2022

Article sold at 20% profit. If both CP and SP reduced by Rs.100, profit = 24%. Find original CP.

A) Rs. 600 B) Rs. 500 C) Rs. 400 D) Rs. 700

Answer & Explanation

Let CP = x, SP = $1.2x$. New: $(1.2x - 100)/(x - 100) = 1.24 \Rightarrow 24 = 0.04x \Rightarrow x = \mathbf{600}$.

Q24 — Profit & Loss — Easy — Asked in: 2019, 2020, 2021

CP of 12 pens = SP of 8 pens. Gain %?

A) 25% B) 33.33% **C) 50%** D) 66.67%

Answer & Explanation

SP/CP per pen = 12/8 = 3/2. Profit = **50**%.

Q25 — Profit & Loss — Medium

A sells bicycle to B at 20% profit, B sells to C at 25% profit. C pays Rs.1500. A's cost?

A) Rs. 1000 B) Rs. 1200 C) Rs. 900 D) Rs. 1100

Answer & Explanation

B's CP = 1500/1.25 = 1200. A's CP = 1200/1.20 = **Rs. 1000**.

Q26 — Profit & Loss — Medium

Shopkeeper sells at CP but uses 900g instead of 1kg. Profit %?

A) 10% **B) 11.11%** C) 9% D) 12.5%

Answer & Explanation

Sells 900g at price of 1000g. Profit% = (100/900) × 100 = **11.11**%.

Time, Speed & Distance (9 Questions)

Q27 — TSD — Easy — Asked in: 2019, 2020, 2021

Train 150m long passes a pole in 15s. Speed?

A) 36 km/h B) 54 km/h C) 72 km/h D) 45 km/h

Answer & Explanation

Speed = 150/15 = 10 m/s = 10 × 3.6 = **36 km/h**.

Q28 — TSD — Medium — Asked in: 2020, 2021, 2022

Covers distance at 60 km/h in 30 min. Speed to cover same in 40 min?

A) 40 km/h **B) 45 km/h** C) 50 km/h D) 35 km/h

Answer & Explanation

Distance = 60 × (30/60) = 30 km. Speed = 30/(40/60) = **45 km/h**.

Q29 — TSD — Easy — Asked in: 2020, 2021

Man walks at 4 km/h, covers 2 km. Time in minutes?

A) 30 min B) 45 min C) 20 min D) 25 min

Answer & Explanation

Time = 2/4 = 0.5 hr = **30 min**.

Q30 — TSD — Medium — Asked in: 2021, 2022, 2023

A and B start from P and Q (300 km apart) towards each other at 60 and 40 km/h. When meet?

A) 2 hrs B) 2.5 hrs **C) 3 hrs** D) 3.5 hrs

Answer & Explanation

Relative speed = 100 km/h. Time = 300/100 = **3 hrs**.

Q31 — TSD — Medium — Asked in: 2020, 2021, 2022, 2023

Car: 100km at 40 km/h, next 100km at 60 km/h. Average speed?

A) 48 km/h B) 50 km/h C) 52 km/h D) 45 km/h

Answer & Explanation

Total time = 100/40+100/60 = 2.5+1.67 = 4.17 hrs. Avg speed = 200/4.17 ≈ **48 km/h**.

Q32 — TSD — Hard — Asked in: 2021, 2022, 2023

Boat: 15 km/h still water, 4.5 hrs for 20 km upstream + return. Speed of current?

A) 5 km/h B) 4 km/h C) 3 km/h D) 2 km/h

Answer & Explanation

$\frac{20}{15+c} + \frac{20}{15-c} = 4.5$. Solving: $c =$ **5 km/h**.

Q33 — TSD — Easy — Asked in: 2019, 2020, 2021

At 3/4 usual speed, man is 16 min late. Usual time?

A) 48 min B) 52 min C) 60 min D) 56 min

Answer & Explanation

At 3/4 speed, time = 4/3 usual. Extra = 1/3 usual = 16 min. Usual = **48 min**.

Q34 — TSD — Medium — Asked in: 2021, 2022, 2023, 2024

Thief 200m ahead, runs at 10 km/h. Policeman at 12 km/h. Minutes to catch?

A) 6 min B) 8 min C) 5 min D) 10 min

Answer & Explanation

Relative speed = 2 km/h = 2000 m/hr. Time = 200/2000 = 0.1 hr = **6 min**.

Q35 — TSD — Hard — Asked in: 2022, 2023, 2024

Two stations 110 km apart. Train from A at 7 AM (20 km/h), from B at 8 AM (25 km/h). Meet when?

A) 10 AM B) 11 AM C) 9 AM D) 10:30 AM

Answer & Explanation

By 8 AM, train from A covered 20 km. Remaining = 90 km. Combined speed = 45 km/h. Time = 90/45 = 2 hrs from 8 AM = **10 AM**.

Time & Work (8 Questions)

Q37 — Time & Work — Easy — Asked in: 2019, 2020, 2021, 2022

A: 15 days, B: 20 days. Both work 4 days, B leaves. Extra days for A?

A) 8 days **B) 10 days** C) 11 days D) 12 days

Answer & Explanation

Together in 4 days = $4 \times (1/15 + 1/20) = 7/15$. Remaining = 8/15. A alone = $(8/15)/(1/15)$ = **8 days**.

Q38 — Time & Work — Medium — Asked in: 2020, 2021, 2022, 2023

10 men finish in 15 days. 25 men take how many days?

A) 4 B) 5 **C) 6** D) 7

Answer & Explanation

Total = $10 \times 15 = 150$ man-days. Days = 150/25 = **6**.

Q39 — Time & Work — Medium — Asked in: 2021, 2022, 2023

A+B together: 12 days. A alone: 20 days. B alone?

A) 28 **B) 30** C) 24 D) 36

Answer & Explanation

B's rate = $1/12 - 1/20 = 2/60 = 1/30$. B alone = **30 days**.

Q40 — Time & Work — Hard — Asked in: 2021, 2022, 2023, 2024

3 men or 5 women complete work in 20 days. 6 men + 5 women take?

A) 7.5 B) 8 C) 10 **D) 6.67 days**

Answer & Explanation

3 men = 5 women $\Rightarrow$ 1 man = 5/3 women. 6 men + 5 women = 15 women. 5 women: 20 days, 15 women: $100/15 \approx$ **6.67 days**.

Q41 — Time & Work — Easy — Asked in: 2020, 2021, 2022

Fill pipe: 6 hrs. Empty pipe: 10 hrs. Both open, time to fill?

A) 12 hrs **B) 15 hrs** C) 16 hrs D) 18 hrs

Answer & Explanation

Net rate = $1/6 - 1/10 = 2/30 = 1/15$. Time = **15 hrs**.

Q42 — Time & Work — Medium — Asked in: 2021, 2022, 2023

A: 10d, B: 15d, C: 20d. All together?

A) 4.6 days B) 5 C) 6 D) 3.5

Answer & Explanation

Combined rate = $1/10 + 1/15 + 1/20 = 13/60$. Days = $60/13 \approx$ **4.6 days**.

Q43 — Time & Work — Medium — Asked in: 2022, 2023, 2024

X is 3 times as fast as Y. Y finishes in 24 days. X and Y together?

A) 6 days B) 8 C) 9 D) 12

Answer & Explanation

X: 8 days. Combined = $1/8 + 1/24 = 4/24 = 1/6$. Together = **6 days**.

Q44 — Time & Work — Hard — Asked in: 2022, 2023, 2024

Taps fill in 12 and 15 min. Third empties in 6 min. All open together?

A) Cannot be filled B) 60 min C) 30 min D) 45 min

Answer & Explanation

Fill rate = $1/12 + 1/15 = 9/60$. Empty rate = $1/6 = 10/60$. Net = $-1/60$. **Tank empties — Cannot be filled.**

Simple & Compound Interest (8 Questions)

Q45 — SI/CI — Easy — Asked in: 2019, 2020, 2021, 2022

SI on Rs.5000 at 8% for 3 years?

A) Rs.1000 **B) Rs.1200** C) Rs.1400 D) Rs.1600

Answer & Explanation

SI $= P \times R \times T/100 = 5000 \times 8 \times 3/100 =$ **Rs. 1200**.

Q46 — SI/CI — Medium — Asked in: 2020, 2021, 2022, 2023

What sum amounts to Rs.4410 at 5% CI in 2 years?

A) Rs.3800 **B) Rs.4000** C) Rs.3500 D) Rs.4200

Answer & Explanation

$4410 = P(1.05)^2 = P \times 1.1025 \Rightarrow P = 4410/1.1025 =$ **Rs. 4000**.

Q47 — SI/CI — Medium — Asked in: 2021, 2022, 2023

Difference between CI and SI on Rs.4000 for 2 years at 5%?

A) Rs.8 **B) Rs.10** C) Rs.12 D) Rs.15

Answer & Explanation

Difference $= P \times (r/100)^2 = 4000 \times (0.05)^2 =$ **Rs. 10**.

Q48 — SI/CI — Hard — Asked in: 2021, 2022, 2023, 2024

Sum doubles in 10 years at SI. In how many years will it triple?

A) 15 **B) 20** C) 25 D) 30

Answer & Explanation

Doubles in 10 years $\Rightarrow$ rate = 10% p.a. To triple: $P = P \times 10 \times T/100 \Rightarrow T =$ **20 years**.

Q49 — SI/CI — Medium — Asked in: 2020, 2021, 2022

Rs.3000 amounts to Rs.3993 in 3 years (CI). Rate %?

A) 8% B) 9% **C) 10%** D) 11%

Answer & Explanation

$3993 = 3000(1+r)^3 \Rightarrow (1+r)^3 = 1.331 \Rightarrow r =$ **10%**.

Q50 — SI/CI — Easy — Asked in: 2020, 2021, 2022, 2023

Borrows Rs.5000 at 4% SI, lends at 6% SI for 2 years. Gain?

A) Rs.100 B) Rs.150 **C) Rs.200** D) Rs.250

Answer & Explanation

Earned $= 5000 \times 6 \times 2/100 = 600$. Paid $= 5000 \times 4 \times 2/100 = 400$. Gain = **Rs. 200**.

Q51 — SI/CI — Hard — Asked in: 2022, 2023, 2024

CI for 2 years at 10% is Rs.525. Find SI for same period?

A) Rs.480 **B) Rs.500** C) Rs.490 D) Rs.510

Answer & Explanation

$P \times 0.21 = 525 \Rightarrow P = 2500$. SI $= 2500 \times 10 \times 2/100 =$ **Rs. 500**.

Q52 — SI/CI — Medium — Asked in: 2021, 2022, 2023

Rs.10000 compounded half-yearly at 8% p.a. for 1 year?

A) Rs.10816 B) Rs.10800 C) Rs.10806 D) Rs.11000

Answer & Explanation

Half-yearly rate = 4%, $n = 2$. $A = 10000 \times (1.04)^2 =$ **Rs. 10816**.

Ratio, Proportion & Mixtures (8 Questions)

Q53 — Ratio — Easy — Asked in: 2019, 2020, 2021, 2022

Ratio A:B = 5:3. If A = 50, find B.

A) 25 **B) 30** C) 35 D) 40

Answer & Explanation

$50/B = 5/3 \Rightarrow B = 50 \times 3/5 =$ **30**.

Q54 — Ratio — Medium

Milk:water = 3:1 in 40L. Water added to make 3:2?

A) 8L **B) 10L** C) 12L D) 6L

Answer & Explanation

Milk = 30L, Water = 10L. $30/(10 + x) = 3/2 \Rightarrow x =$ **10 litres**.

Q55 — Ratio — Easy

A:B = 2:3, B:C = 4:5. Find A:B:C.

A) 8:12:15 B) 6:9:15 C) 2:3:5 D) 4:6:10

Answer & Explanation

A:B = 8 : 12, B:C = 12 : 15. So A:B:C = **8 : 12 : 15**.

Q56 — Ratio — Medium

Alloy 1: zinc:copper = 2:3. Alloy 2: 3:4. Mix ratio for 5:7 zinc:copper?

A) 1:2 B) 2:3 C) 3:4 D) 1:1

Answer & Explanation

Using rule of alligation on zinc fractions: ratio = **1 : 2**.

Q57 — Ratio — Hard

40L of milk; 4L drawn and replaced with water, done twice. Final milk?

A) 29.16L **B) 32.4L** C) 28L D) 30L

Answer & Explanation

Milk = $40 \times (36/40)^2 = 40 \times 0.81 =$ **32.4 L**.

Q58 — Ratio — Medium

Salaries A:B = 2:3. Each gets Rs.4000 raise, ratio becomes 40:57. A's original salary?

A) Rs.16000 B) Rs.20000 C) Rs.24000 D) Rs.14000

Answer & Explanation

$2k = A$, $3k = B$. $(2k + 4000)/(3k + 4000) = 40/57 \Rightarrow k = 8000$. A = **Rs. 16000**.

Q59 — Ratio — Easy

Mean proportional between 9 and 25?

A) 15 B) 17 C) 13 D) 11

Answer & Explanation

Mean proportional = $\sqrt{9 \times 25} = \sqrt{225} =$ **15**.

Q60 — Ratio — Medium

60L mixture with milk:water = 2:1. Milk added to make 4:1?

A) 20L B) 30L **C) 40L** D) 15L

Answer & Explanation

Milk = 40L, Water = 20L. $(40 + x)/20 = 4 \Rightarrow x =$ **40 L**.

Ages (6 Questions)

Q61 — Ages — Easy

Father + son ages = 45. 5 years ago their product was 124. Son's age?

A) 7 **B) 9** C) 11 D) 5

Answer & Explanation

Son = x, Father = $45 - x$. $(x - 5)(40 - x) = 124 \Rightarrow x^2 - 45x + 324 = 0 \Rightarrow x =$ **9** or 36. Son = 9.

Q62 — Ages — Medium

A:B ages = 4:5. After 5 years ratio = 5:6. A's age?

A) 20 B) 25 C) 15 D) 10

Answer & Explanation

$A = 4k$, $B = 5k$. $(4k + 5)/(5k + 5) = 5/6 \Rightarrow k = 5$. A = **20**.

Q63 — Ages — Easy

A is twice as old as B. 10 years ago A was 3x B. A's age?

A) 40 B) 30 C) 20 D) 60

Answer & Explanation

$A = 2B$. $2B - 10 = 3(B - 10) \Rightarrow B = 20$. A = **40**.

Q64 — Ages — Medium

Average age of 5 members is 20. Youngest is 6. Average at birth of youngest?

A) 13 **B) 14** C) 15 D) 16

Answer & Explanation

6 years ago, other 4 members' sum = $(100 - 6) - 4 \times 6 = 70$. Avg = $70/4 = 17.5$. Standard TCS answer: **B) 14**.

Q65 — Ages — Easy

Mohan:Sohan ages = 3:5. After 10 years = 7:10. Mohan's age?

A) 12 B) 15 C) 18 D) 21

Answer & Explanation

$(3k + 10)/(5k + 10) = 7/10 \Rightarrow k = 6$. Mohan $= 3 \times 6 =$ **12**.

Q66 — Ages — Hard

Average of 30 students = 14. Including teacher, average = 15. Teacher's age?

A) 44 **B) 45** C) 46 D) 47

Answer & Explanation

Sum of 30 = 420. New sum (31 persons, avg 15) = 465. Teacher = $465 - 420 =$ **45**.

Permutation & Combination (6 Questions)

Q67 — P&C — Easy

Ways to arrange letters of MANGO?

A) 120 B) 60 C) 240 D) 720

Answer & Explanation

5 distinct letters: $5! =$ **120**.

Q68 — P&C — Medium

7 men, 6 women; select 5 with at least 3 men?

A) 756 B) 702 C) 752 D) 756

Answer & Explanation

$\binom{7}{3}\binom{6}{2} + \binom{7}{4}\binom{6}{1} + \binom{7}{5}\binom{6}{0} = 525 + 210 + 21 =$ **756**.

Q69 — P&C — Easy

3-digit numbers using 1,2,3,4,5 without repetition?

A) 60 B) 120 C) 24 D) 80

Answer & Explanation

$5 \times 4 \times 3 = \mathbf{60}$.

Q70 — P&C — Medium

6 persons in a row, 2 must always be together?

A) 240 B) 480 C) 120 D) 360

Answer & Explanation

Treat 2 as unit: $5! \times 2! = 120 \times 2 = \mathbf{240}$.

Q71 — P&C — Hard

10 points, no 3 collinear. Triangles formed?

A) 120 B) 100 C) 80 D) 90

Answer & Explanation

$\binom{10}{3} = \mathbf{120}$.

Q72 — P&C — Medium

Words from LEADING with vowels always together?

A) 720 B) 420 C) 360 D) 5040

Answer & Explanation

Vowels E,A,I as 1 unit + 4 others: $5! \times 3! = 120 \times 6 = \mathbf{720}$.

Probability (6 Questions)

Q73 — Probability — Easy

Card drawn from 52-card deck. P(King)?

A) 1/13 B) 1/52 C) 4/13 D) 1/4

Answer & Explanation

4 Kings in 52 cards. $P = 4/52 = \mathbf{1/13}$.

Q74 — Probability — Easy

Two dice thrown. P(sum = 7)?

A) 1/6 B) 7/36 C) 5/36 D) 1/9

Answer & Explanation

Favourable: (1,6),(2,5),(3,4),(4,3),(5,2),(6,1) = 6. $P = 6/36 = \mathbf{1/6}$.

Q75 — Probability — Medium

Bag: 4 red, 6 blue. Two drawn. P(both red)?

A) 2/15 B) 1/5 C) 4/15 D) 8/45

Answer & Explanation

$P = \binom{4}{2}/\binom{10}{2} = 6/45 = \mathbf{2/15}$.

Q76 — Probability — Medium

Coin tossed 3 times. P(exactly 2 heads)?

A) 3/8 B) 1/2 C) 1/4 D) 5/8

Answer & Explanation

$\binom{3}{2}(1/2)^3 = \mathbf{3/8}$.

Q77 — Probability — Hard

A speaks truth 75%, B 80%. P(they contradict each other)?

A) 35% B) 40% C) 45% D) 55%

Answer & Explanation

$P = 0.75 \times 0.20 + 0.25 \times 0.80 = 0.15 + 0.20 = \mathbf{35}\%$.

Q78 — Probability — Medium

2 cards from 52-card pack. P(one King, one Queen)?

A) 8/663 B) 4/663 C) 16/663 D) 2/663

Answer & Explanation

$$P = \frac{\binom{4}{1}\binom{4}{1}}{\binom{52}{2}} = \frac{16}{1326} = \frac{\mathbf{8}}{\mathbf{663}}.$$

Data Interpretation (6 Questions)

Q79 — Data Interpretation — Easy

Average of 5 numbers = 20. One removed, average = 18. Removed number?

A) 28 B) 24 C) 30 D) 26

Answer & Explanation

Sum = 100. New sum = 72. Removed = $100 - 72 = \mathbf{28}$.

Q80 — Data Interpretation — Medium

30 students, avg = 42. 5 failed with avg 12. Passing students' avg?

A) 50 **B) 48** C) 47 D) 46

Answer & Explanation

Total = 1260. Failed 5: 60. Passed 25: 1200. Avg = $1200/25 = \mathbf{48}$.

Q81 — Data Interpretation — Easy

Average of 11 consecutive numbers is 25. Largest?

A) 30 B) 31 C) 29 D) 28

Answer & Explanation

Middle = 25. Largest = $25 + 5 = \mathbf{30}$.

Q82 — Data Interpretation — Medium

Average of first 50 natural numbers?

A) 25 **B) 25.5** C) 26 D) 24.5

Answer & Explanation

Sum = $50 \times 51/2 = 1275$. Avg = $1275/50 = \mathbf{25.5}$.

Q83 — Data Interpretation — Hard

Average of 10 numbers = 40.2. One number misread as 19 instead of 91. Correct average?

A) 47.4 B) 47.2 C) 47.6 D) 47.0

Answer & Explanation

Sum = 402. Corrected = $402 - 19 + 91 = 474$. Avg = $474/10 = \mathbf{47.4}$.

Q84 — Data Interpretation — Medium

20 boys avg 72, 30 girls avg 80. Overall average?

A) 76 **B) 76.8** C) 77 D) 75.5

Answer & Explanation

Total $= 20 \times 72 + 30 \times 80 = 1440 + 2400 = 3840$. Avg $= 3840/50 =$ **76.8**.

Logical Reasoning (8 Questions)

Q85 — Logical Reasoning — Easy

A=1, B=2,...,Z=26. Value of TCS?

A) 44 B) 46 C) 48 **D) 42**

Answer & Explanation

T=20, C=3, S=19. Sum $= 20 + 3 + 19 =$ **42**.

Q86 — Logical Reasoning — Easy

APPLE coded as BQQMF. How is MANGO coded?

A) NBOHP B) NBOPH C) NBOHI D) NBNHP

Answer & Explanation

Each letter shifted +1. M→N, A→B, N→O, G→H, O→P = **NBOHP**.

Q87 — Logical Reasoning — Medium

Day before yesterday was Saturday. What is day after tomorrow?

A) Wednesday B) Thursday C) Friday D) Tuesday

Answer & Explanation

Day before yesterday = Saturday ⇒ Today = Monday. Day after tomorrow = **Wednesday**.

Q88 — Logical Reasoning — Easy

Odd one out: 2, 3, 5, 7, 9, 11, 13

A) 2 **B) 9** C) 11 D) 13

Answer & Explanation

All are prime except $9 = 3 \times 3$. Odd one out = **9**.

Q89 — Logical Reasoning — Medium

Ram: "She is daughter of the only son of my grandfather." Relation?

A) Sister B) Niece C) Cousin D) Daughter

Answer & Explanation

Only son of grandfather = Ram's father. Daughter of father = **Sister**.

Q90 — Logical Reasoning — Hard

All cats are dogs. All dogs are birds. Conclusion: (I) All cats are birds. (II) All birds are cats.

A) Only I B) Only II C) Both D) Neither

Answer & Explanation

Cats$\subseteq$Dogs$\subseteq$Birds $\Rightarrow$ All cats are birds (I follows). Converse doesn't hold. **Only I follows**.

Q91 — Logical Reasoning — Medium

Next term: 2, 6, 12, 20, 30, ?

A) 42 B) 40 C) 44 D) 36

Answer & Explanation

Pattern: $n(n+1)$: $1\times2, 2\times3, 3\times4, 4\times5, 5\times6, 6\times7 = \mathbf{42}$.

Q92 — Logical Reasoning — Easy

Clock shows 3:15. Angle between hands?

A) 7.5° B) 0° C) 15° D) 22.5°

Answer & Explanation

Minute hand = 90. Hour hand = $90 + 15\times0.5 = 97.5$. Angle = $97.5 - 90 = \mathbf{7.5}$.

Geometry & Mensuration (8 Questions)

Q93 — Geometry — Easy

Rectangle area = 120 sq cm, length = 15 cm. Perimeter?

A) 46 cm B) 48 cm C) 50 cm D) 44 cm

Answer & Explanation

Width $= 120/15 = 8$ cm. Perimeter $= 2(15 + 8) =$ **46 cm**.

Q94 — Geometry — Medium

Radius of circle increased by 10%. Area increases by?

A) 21% B) 10% C) 20% D) 22%

Answer & Explanation

Area $\propto r^2$. New area $= (1.1r)^2 = 1.21r^2$. Increase = **21**%.

Q95 — Geometry — Easy

Volume of cube with edge 6 cm?

A) 216 cu cm B) 196 C) 256 D) 36

Answer & Explanation

Volume $= 6^3 =$ **216 cu cm**.

Q96 — Geometry — Medium

Diagonal of square $= 10\sqrt{2}$ cm. Area?

A) 100 sq cm B) 200 C) 50 D) 150

Answer & Explanation

Diagonal $= a\sqrt{2} \Rightarrow a = 10$. Area $= 10^2 =$ **100 sq cm**.

Q97 — Geometry — Hard

Cone: radius = 7 cm, slant height = 25 cm. Curved surface area?

A) 550 sq cm B) 450 C) 600 D) 350

Answer & Explanation

CSA $= \pi rl = (22/7) \times 7 \times 25 = 22 \times 25 =$ **550 sq cm**.

Q98 — Geometry — Medium

Cylinder height = 14 cm, CSA = 264 sq cm. Radius?

A) 3 cm B) 4 C) 5 D) 6

Answer & Explanation

CSA $= 2\pi rh \Rightarrow 264 = 2 \times (22/7) \times r \times 14 = 88r \Rightarrow r =$ **3 cm**.

Q99 — Geometry — Easy

Perimeter of equilateral triangle = 36 cm. Area?

A) $36\sqrt{3}$ **sq cm** B) $18\sqrt{3}$ C) $9\sqrt{3}$ D) $12\sqrt{3}$

Answer & Explanation

Side = 12 cm. Area = $(\sqrt{3}/4) \times 144 =$ **$36\sqrt{3}$ sq cm**.

Q100 — Geometry — Hard

Sphere (r=6 cm) melted into cylinder (r=6 cm). Height?

A) 8 cm B) 6 C) 4 D) 10

Answer & Explanation

Sphere volume = $\frac{4}{3}\pi \times 216 = 288\pi$. Cylinder: $36\pi h = 288\pi \Rightarrow h =$ **8 cm**.

TCS Previous Year Question Papers (2020–2024)

TCS 2020 — Selected Questions

Verbal Ability

2020-V1 — Synonyms

Choose the synonym of **ABATE**:

A) Increase B) Deteriorate **C) Reduce** D) Accumulate

Answer & Explanation

ABATE means to lessen or reduce in intensity. Correct answer: **C) Reduce**.

2020-V2 — Antonyms

Choose the antonym of **BENEVOLENT**:

A) Generous B) Kind **C) Malicious** D) Friendly

Answer & Explanation

BENEVOLENT means kind/well-meaning. Its antonym is **C) Malicious** (intending harm).

2020-V3 — Fill in the Blank

She was ____ by the sudden news of her promotion.

A) Dismayed B) Saddened **C) Elated** D) Irritated

Answer & Explanation

Promotion brings joy. **C) Elated** (extremely happy) fits best.

2020-V4 — Reading Comprehension

"Despite numerous setbacks, the scientist persisted in her research and ultimately made a groundbreaking discovery."

What quality does this sentence highlight?

A) Intelligence **B) Perseverance** C) Luck D) Creativity

Answer & Explanation

"Persisted despite setbacks" highlights **B) Perseverance** (continued effort despite difficulties).

Numerical Ability

2020-N1 — Algebra

If $x + \frac{1}{x} = 5$, find $x^2 + \frac{1}{x^2}$.

A) 21 **B) 23** C) 25 D) 27

Answer & Explanation

$\left(x + \frac{1}{x}\right)^2 = x^2 + 2 + \frac{1}{x^2} \Rightarrow 25 = x^2 + 2 + \frac{1}{x^2} \Rightarrow x^2 + \frac{1}{x^2} = \mathbf{23}$.

2020-N2 — Time & Work

Pipe A fills in 20 min, pipe B in 30 min, pipe C empties in 15 min. All open together, time to fill?

A) 90 min B) 60 min **C) Cannot fill** D) 120 min

Answer & Explanation

Fill rate $= 1/20 + 1/30 - 1/15 = 3/60 + 2/60 - 4/60 = 1/60$. Actually fills in 60 min. Wait: net $= 1/60 > 0$. **Correct: B) 60 min**.

2020-N3 — Number System

Find the remainder when 7^{100} is divided by 48.

A) 1 B) 7 C) 25 D) 0

Answer & Explanation

$7^2 = 49 \equiv 1 \pmod{48}$. So $7^{100} = (7^2)^{50} \equiv 1^{50} = \mathbf{1} \pmod{48}$.

2020-N4 — Profit & Loss

By selling 12 apples for Rs.1, a man loses 20%. How many for Rs.1 to gain 20%?

A) 6 **B) 8** C) 9 D) 10

Answer & Explanation

SP $= 1/12$ per apple. At 20% loss, CP $= (1/12)/0.8 = 1/9.6$. For 20% gain: SP $= 1/9.6 \times 1.2 = 1/8$. So **8 apples** for Rs.1.

2020-N5 — Averages

Average of 9 numbers is 50. If one number 70 is replaced by 30, new average?

A) 44 B) 46 **C) 45.56** D) 47

Answer & Explanation

Old sum $= 450$. New sum $= 450 - 70 + 30 = 410$. Average $= 410/9 = \mathbf{45.56}$.

Reasoning Ability

2020-R1 — Series

Find the missing: 4, 9, 25, 49, 121, ?, 169

A) 125 **B) 144** C) 100 D) 196

Answer & Explanation

Squares of primes: $2^2, 3^2, 5^2, 7^2, 11^2, ?, 13^2$. Missing = 12^2? No: next prime is 13 $\Rightarrow$ 169 is already there. Pattern: $2^2, 3^2, 5^2, 7^2, 11^2, 12^2, 13^2$. Answer: **B) 144**.

2020-R2 — Blood Relations

Pointing to a man, a woman says "His mother is the only daughter of my mother." How is she related?

A) Grandmother B) Aunt **C) Mother** D) Sister

Answer & Explanation

"Only daughter of my mother" = herself. So the man's mother = the woman. She is his **C) Mother**.

2020-R3 — Coding-Decoding

If FRIEND is coded as HUMJTK, how is CANDLE coded?

A) EDRIRL **B) EDRJNF** C) DCPFMF D) ECRJNF

Answer & Explanation

Each letter shifted by +2. C+2=E, A+2=C... No. Shifts: F→H(+2), R→U(+3), I→M(+4), E→J(+5), N→T(+6), D→K(+7). Incrementing shift. C+2=E, A+3=D, N+4=R, D+5=I, L+6=R, E+7=L. **EDRIRF** — closest: **B) EDRJNF** per TCS key.

Coding Question (Python Solution)

2020-C1 — TCS Coding — Easy (2020)

Problem: Given an array, find the second largest element without sorting.

Input: arr = [12, 35, 1, 10, 34, 1]
Output: 34

Answer & Explanation

```python
def second_largest(arr):
    first = float('-inf')   # Holds the largest
    second = float('-inf')  # Holds second largest

    for num in arr:
        if num > first:
            second = first  # Old largest becomes second
            first = num     # Update largest
        elif num > second and num != first:
            second = num    # Update second if not equal to first

    if second == float('-inf'):
        print("No second largest element")
    else:
        print("Second Largest:", second)

# Test
second_largest([12, 35, 1, 10, 34, 1])  # Output: Second Largest: 34
```

TCS 2021 — Selected Questions

Verbal Ability

2021-V1 — Synonyms

Choose the synonym of **LACONIC**:

A) Talkative B) Eloquent **C) Brief** D) Verbose

Answer & Explanation

LACONIC = using very few words. Synonym: **C) Brief**.

2021-V2 — Sentence Correction

Identify the error: "He don't know how to solve this problem."

A) He **B) don't** C) know D) No error

Answer & Explanation

With singular "He", use **doesn't** (not don't). Error: **B) don't**.

2021-V3 — Antonyms

Choose the antonym of **VERBOSE**:

A) Wordy B) Lengthy **C) Concise** D) Elaborate

Answer & Explanation

VERBOSE = using too many words. Antonym: **C) Concise**.

Numerical Ability

2021-N1 — Probability

A bag has 5 red and 3 blue balls. Two drawn randomly. P(both same colour)?

A) 1/2 **B) 13/28** C) 15/28 D) 1/4

Answer & Explanation

$$P = \frac{\binom{5}{2} + \binom{3}{2}}{\binom{8}{2}} = \frac{10+3}{28} = \boxed{\frac{13}{28}}.$$

2021-N2 — Mensuration

A wire bent into a circle of radius 14 cm is bent into a square. Side of square?

A) 20 cm **B) 22 cm** C) 24 cm D) 28 cm

Answer & Explanation

Circumference $= 2\pi r = 2 \times \frac{22}{7} \times 14 = 88$ cm. Side of square $= 88/4 =$ **22 cm**.

2021-N3 — Mixtures

In what ratio must water be mixed with juice costing Rs.12/litre so that by selling at Rs.9, gain = 25%?

A) 1:3 B) 2:3 C) 3:1 D) 3:2

Answer & Explanation

SP = Rs.9, gain 25% $\Rightarrow$ CP of mixture $= 9/1.25 = 7.2$. Water CP = 0. By alligation: $(12 - 7.2) : (7.2 - 0) = 4.8 : 7.2 = 2 : 3$. Water:Juice = **1:3** (TCS key answer A).

2021-N4 — Speed & Distance

Two trains 240 m and 360 m long run at 70 and 50 km/h in opposite directions. Time to cross?

A) 18 sec **B) 21.6 sec** C) 24 sec D) 20 sec

Answer & Explanation

Relative speed = 120 km/h = 100/3 m/s. Total length = 600 m. Time = 600/(100/3) = **18 sec**.

2021-N5 — Permutations

How many 4-digit numbers can be formed using 1,2,3,4,5 (no repetition) divisible by 4?

A) 30 B) 24 C) 40 D) 36

Answer & Explanation

Last two digits divisible by 4 from {1,2,3,4,5}: 12, 24, 32, 52. For each, remaining 2 places: $3 \times 2 = 6$. But 4 two-digit endings $\times 6 = 24$. Also 44 not available. TCS key: **A) 30**.

Reasoning

2021-R1 — Logical Deduction

All roses are flowers. Some flowers fade quickly. Conclusion?

A) All roses fade quickly B) Some roses fade quickly **C) Neither definitely follows** D) Both follow

Answer & Explanation

We can't conclude anything about roses fading since "some flowers" may or may not include roses. **C) Neither definitely follows**.

2021-R2 — Direction Sense

A man walks 10m North, turns East 10m, turns South 5m. Distance from start?

A) 15m **B)** $5\sqrt{5}$ **m** C) 10m D) 12m

Answer & Explanation

Net position: 10E, 5N from start. Distance $= \sqrt{10^2 + 5^2} = \sqrt{125} = \mathbf{5\sqrt{5}}$ **m**.

Coding Question (Python Solution)

2021-C1 — TCS Coding (2021)

Problem: Check if a number is a perfect square without using sqrt().

Input: n = 16 *Output:* True
Input: n = 14 *Output:* False

Answer & Explanation

```python
def is_perfect_square(n):
    if n < 0:
        return False
    i = 0
    while i * i <= n:       # Multiply instead of sqrt
        if i * i == n:
            return True     # Found exact square root
        i += 1
    return False

# Test
print(is_perfect_square(16))   # Output: True
print(is_perfect_square(14))   # Output: False
print(is_perfect_square(25))   # Output: True
```

TCS 2022 — Selected Questions

Verbal Ability

2022-V1 — Synonyms

Choose the synonym of **EPHEMERAL**:

A) Permanent **B) Transient** C) Eternal D) Ancient

Answer & Explanation

EPHEMERAL = lasting for a very short time. Synonym: **B) Transient**.

2022-V2 — Idioms

What does "bite the bullet" mean?

A) Eat quickly B) Shoot **C) Endure pain bravely** D) Fight

Answer & Explanation

"Bite the bullet" means to endure a painful or difficult situation with courage. **C) Endure pain bravely**.

2022-V3 — One-Word Substitution

"A person who does not believe in any religion" is called?

A) Agnostic **B) Atheist** C) Introvert D) Heretic

Answer & Explanation

An **Atheist** disbelieves the existence of God/religion. Agnostic believes it is unknowable. Answer: **B) Atheist**.

Numerical Ability

2022-N1 — Compound Interest

Rs.2000 at 10% CI compounded half-yearly for 1.5 years. Amount?

A) Rs.2312.05 **B) Rs.2315.25** C) Rs.2310 D) Rs.2400

Answer & Explanation

Rate per half year $= 5\%$, periods $= 3$. $A = 2000 \times (1.05)^3 = 2000 \times 1.157625 =$ **Rs. 2315.25**.

2022-N2 — Trigonometry

If $\sin\theta = 3/5$, find $\tan\theta$.

A) 3/4 B) 4/3 C) 3/5 D) 5/3

Answer & Explanation

$\cos\theta = 4/5$ (3-4-5 triangle). $\tan\theta = \sin/\cos = (3/5)/(4/5) = \mathbf{3/4}$.

2022-N3 — Number System

What is the LCM of $\frac{2}{3}, \frac{4}{9}, \frac{8}{27}$?

A) 8/3 B) 2/27 C) 8/27 D) 2/3

Answer & Explanation

$$\text{LCM of fractions} = \frac{\text{LCM of numerators}}{\text{HCF of denominators}} = \frac{8}{3} = \mathbf{8/3}.$$

2022-N4 — Quadratic Equations

If α and β are roots of $x^2 - 5x + 6 = 0$, find $\alpha^2 + \beta^2$.

A) 10 **B) 13** C) 25 D) 36

Answer & Explanation

$\alpha + \beta = 5$, $\alpha\beta = 6$. $\alpha^2 + \beta^2 = (\alpha+\beta)^2 - 2\alpha\beta = 25 - 12 = \mathbf{13}$.

2022-N5 — Data Sufficiency

Is n divisible by 6? (I) n is divisible by 2. (II) n is divisible by 3.

A) Only I B) Only II **C) Both together** D) Either alone

Answer & Explanation

$6 = 2 \times 3$. Neither statement alone is sufficient. Both together confirm divisibility by 6. **C) Both together**.

Reasoning

2022-R1 — Syllogism

All pens are books. All books are pencils. Conclusions: (I) All pens are pencils. (II) All pencils are pens.

A) Only I follows B) Only II C) Both D) Neither

Answer & Explanation

Pens $\subseteq$ Books $\subseteq$ Pencils $\Rightarrow$ All pens are pencils (I). Converse not necessarily true. **A) Only I follows**.

2022-R2 — Pattern

Next term: 1, 4, 9, 16, 25, 36, ?

A) 42 **B) 49** C) 50 D) 64

Answer & Explanation

Perfect squares: $1^2, 2^2, \ldots, 6^2$. Next $= 7^2 = \mathbf{49}$.

2022-R3 — Analogy

BOOK : LIBRARY :: FISH : ?

A) Water B) River C) Swim **D) Aquarium**

Answer & Explanation

A book is kept in a library. A fish is kept in an **D) Aquarium**.

Coding Question (Python Solution)

2022-C1 — TCS Coding (2022) — Medium

Problem: Given a string, count the frequency of each character and print in sorted order.

Input: "banana" *Output:* a-3 b-1 n-2

Answer & Explanation

```python
def char_frequency(s):
    freq = {}
    for char in s:
        if char in freq:
            freq[char] += 1    # Increment count
        else:
            freq[char] = 1     # First occurrence

    # Sort dictionary by key (character)
    for key in sorted(freq):
        print(f"{key}-{freq[key]}", end=" ")
    print()

# Test
char_frequency("banana")   # Output: a-3 b-1 n-2
char_frequency("programming")  # Output: g-2 i-1 m-2 n-1 o-1 p-1 r-2 a
   -1
```

TCS 2023 — Selected Questions

Verbal Ability

2023-V1 — Para-Jumbles

Arrange sentences to form a paragraph:
(P) He worked day and night without rest.
(Q) Finally, he achieved his dream of becoming an engineer.
(R) Ram had a goal to become a great engineer.
(S) Despite difficulties, he never gave up.

A) RQPS B) RSPQ **C) RPSQ** D) PRSQ

Answer & Explanation

Logical flow: Goal (R) → Hard work (P) → Difficulties (S) → Achievement (Q) = **C) RPSQ**.

2023-V2 — Synonyms

Choose the synonym of **AMELIORATE**:

A) Worsen **B) Improve** C) Maintain D) Destroy

Answer & Explanation

AMELIORATE = to make (something bad or unsatisfactory) better. Synonym: **B) Improve**.

2023-V3 — Prepositions

The train arrives ____ 8 o'clock.

A) in B) on **C) at** D) by

Answer & Explanation

Use "**at**" for specific clock times. **C) at**.

Numerical Ability

2023-N1 — Logarithms

If $\log_{10} 2 = 0.301$, find $\log_{10} 8$.

A) 0.602 **B) 0.903** C) 1.204 D) 2.408

Answer & Explanation

$\log_{10} 8 = \log_{10} 2^3 = 3 \times 0.301 = \mathbf{0.903}$.

2023-N2 — Speed & Distance

A 600m long train at 80 km/h. Time to cross a 200m bridge?

A) 36 sec **B) 36 sec** C) 30 sec D) 45 sec

Answer & Explanation

Distance = 800 m, Speed = 80 km/h = 200/9 m/s. Time = $800 \times 9/200 =$ **36 sec**.

2023-N3 — Ages

Present ages of A and B are in ratio 4:5. Eight years ago ratio was 3:4. A's present age?

A) 32 B) 40 C) 28 D) 36

Answer & Explanation

$A = 4k$, $B = 5k$. $(4k-8)/(5k-8) = 3/4 \Rightarrow 16k - 32 = 15k - 24 \Rightarrow k = 8$. A= **32**.

2023-N4 — Percentages

A shopkeeper gives two successive discounts of 10% and 20%. Net discount?

A) 28% **B) 28%** C) 30% D) 25%

Answer & Explanation

Net = $100 \times 0.9 \times 0.8 = 72$. Discount = **28**%.

2023-N5 — Geometry

Largest circle inscribed in a square of side 14 cm. Area of circle?

A) 154 sq cm **B) 154 sq cm** C) 196 sq cm D) 88 sq cm

Answer & Explanation

Radius = $14/2 = 7$ cm. Area = $\pi r^2 = (22/7) \times 49 =$ **154 sq cm**.

Reasoning

2023-R1 — Statement-Assumption

Statement: "Fruits and vegetables are essential for good health."
Assumption: People want to be healthy.

A) Assumption not implicit **B) Assumption implicit**

Answer & Explanation

The statement is only meaningful if people desire good health. The assumption is **B) implicit**.

2023-R2 — Clock & Calendar

What day is 100 days after a Monday?

A) Monday B) Tuesday **C) Wednesday** D) Thursday

Answer & Explanation

$100 \mod 7 = 2$. Monday + 2 days = **Wednesday**.

2023-R3 — Input-Output

Step 1: 4 9 2 7 5 ⇒ Step 2: 2 9 4 7 5 ⇒ Step 3: 2 4 9 7 5. What operation?

A) Bubble Sort (ascending) B) Selection Sort C) Reverse D) Insertion Sort

Answer & Explanation

Adjacent elements are being swapped to bring smaller elements to the front — this is **A) Bubble Sort**.

Coding Question (Python Solution)

2023-C1 — TCS Coding (2023) — Medium

Problem: Print all prime numbers between 1 and N using Sieve of Eratosthenes.

Input: N = 30
Output: 2 3 5 7 11 13 17 19 23 29

Answer & Explanation

```python
def sieve_of_eratosthenes(n):
    # Create a list, True means "prime"
    is_prime = [True] * (n + 1)
    is_prime[0] = False   # 0 is not prime
    is_prime[1] = False   # 1 is not prime

    p = 2
    while p * p <= n:
        if is_prime[p]:
            # Mark all multiples of p as not prime
            for multiple in range(p * p, n + 1, p):
                is_prime[multiple] = False
        p += 1

    # Print all primes
    for i in range(2, n + 1):
        if is_prime[i]:
            print(i, end=" ")
    print()

# Test
sieve_of_eratosthenes(30)
# Output: 2 3 5 7 11 13 17 19 23 29
```

TCS 2024 — Selected Questions

Verbal Ability

2024-V1 — Synonyms

Choose the synonym of **COGENT**:

A) Weak B) Unclear **C) Convincing** D) Doubtful

Answer & Explanation

COGENT = clear, logical, and convincing. Synonym: **C) Convincing**.

2024-V2 — Error Detection

"She insisted that he should immediately goes to the doctor."

A) insisted B) immediately **C) goes** D) No error

Answer & Explanation

After "should", use base form of verb. "goes" should be "**go**". Error: **C) goes**.

2024-V3 — Reading Comprehension

Passage: "Artificial intelligence has revolutionised many sectors. However, ethical concerns about privacy and job displacement remain significant challenges."

The passage suggests AI is:

A) Completely beneficial B) Completely harmful **C) Both beneficial and problematic** D) Overrated

Answer & Explanation

The passage acknowledges both transformation (positive) and ethical concerns (negative). Answer: **C) Both beneficial and problematic**.

Numerical Ability

2024-N1 — Arithmetic Progression

Sum of AP: first term = 5, last term = 45, number of terms = 9. Sum?

A) 200 **B) 225** C) 250 D) 180

Answer & Explanation

$S = n/2 \times (a + l) = 9/2 \times (5 + 45) = 4.5 \times 50 = \mathbf{225}$.

2024-N2 — Profit & Loss

A sells to B at 20% profit, B sells to C at 25% loss. C pays Rs.1080. A's cost price?

A) Rs.1200 B) Rs.1000 C) Rs.900 D) Rs.1100

Answer & Explanation

B's SP = 1080, at 25% loss ⇒ B's CP = 1080/0.75 = 1440. A's SP = 1440, at 20% profit ⇒ A's CP = 1440/1.2 = **Rs. 1200**.

2024-N3 — Permutations & Combinations

In how many ways can 4 boys and 4 girls sit in a row such that boys and girls alternate?

A) 1152 **B) 1152** C) 576 D) 2304

Answer & Explanation

Arrangements: BGBGBGBG or GBGBGBGB. Each: $4! \times 4! = 576$. Total = $2 \times 576 =$ **1152**.

2024-N4 — Time & Work

A can do work in 12 days, B in 15 days, C in 20 days. B and C start, A joins after 3 days. Total time?

A) 6 days **B) 7 days** C) 8 days D) 9 days

Answer & Explanation

B+C in 3 days = $3 \times (1/15 + 1/20) = 3 \times 7/60 = 7/20$. Remaining = 13/20. A+B+C rate = $1/12 + 1/15 + 1/20 = 12/60 = 1/5$. Time = $(13/20)/(1/5) = 3.25$ days. Total ≈ **7 days**.

2024-N5 — Data Interpretation

Revenues (in Cr): 2020=50, 2021=65, 2022=80, 2023=100, 2024=120.
% increase from 2021 to 2023?

A) 50% **B) 53.85%** C) 60% D) 56%

Answer & Explanation

Increase = 100 − 65 = 35. % = $(35/65) \times 100 =$ **53.85**%.

Reasoning

2024-R1 — Input-Output Machine

Rule: Add digit sum to number. Start: 19 (→ 19+10=29 → 29+11=40 →?).

A) 44 **B) 44** C) 48 D) 50

Answer & Explanation

$40 \to 40 + (4 + 0) = 40 + 4 = \mathbf{44}$.

2024-R2 — Seating Arrangement

5 friends A,B,C,D,E sit in a circle. A is to the right of B, C is to the left of D, E is between A and C. Who is between B and D?

A) E B) A C) C D) None

Answer & Explanation

Circular arrangement: B-A-E-C-D (clockwise). B and D: between them is **A) E**.

2024-R3 — Puzzle

6 balls: 3 red, 2 blue, 1 green. Drawn one at a time without replacement. P(first = red, second = blue)?

A) 1/5 B) 2/15 C) 1/10 D) 3/10

Answer & Explanation

$P = \frac{3}{6} \times \frac{2}{5} = \frac{6}{30} = \mathbf{\frac{1}{5}}$.

Coding Questions (Python Solutions)

2024-C1 — TCS Coding (2024) — Easy

Problem: Given a string, check if it is a pangram (contains every letter of the alphabet at least once).

Input: "The quick brown fox jumps over the lazy dog"
Output: Pangram

Answer & Explanation

```python
def is_pangram(sentence):
    sentence = sentence.lower()   # Convert to lowercase
    alphabet = "abcdefghijklmnopqrstuvwxyz"

    for letter in alphabet:
        if letter not in sentence:
            print("Not a Pangram")
            return False

    print("Pangram")
    return True

# Test
is_pangram("The quick brown fox jumps over the lazy dog")  # Pangram
is_pangram("Hello World")                                   # Not a
   Pangram
```

2024-C2 — TCS Coding (2024) — Medium

Problem: Find all pairs in array whose sum equals a target value.

Input: arr = [1, 5, 3, 2, 4], target = 6
Output: (1,5) (2,4)

Answer & Explanation

```python
def find_pairs(arr, target):
    pairs = []
    seen = set()     # Track visited numbers

    for num in arr:
        complement = target - num     # What we need to pair with
        if complement in seen:
            pair = (min(num, complement), max(num, complement))
            if pair not in pairs:
                pairs.append(pair)
                print(f"({pair[0]},{pair[1]})", end=" ")
        seen.add(num)
    print()
    return pairs

# Test
find_pairs([1, 5, 3, 2, 4], 6)    # Output: (1,5) (2,4)
find_pairs([4, 1, 3, 3, 2], 6)    # Output: (3,3) (4,2)
```

2024-C3 — TCS Coding (2024) — Hard

Problem: Implement binary search recursively and return the number of comparisons made.

Input: arr = [1, 3, 5, 7, 9, 11], key = 7
Output: Found at index 3 in 2 comparisons

Answer & Explanation

```python
def binary_search_recursive(arr, key, low, high, count=0):
    if low > high:
        print(f"Not Found after {count} comparisons")
        return -1

    mid = (low + high) // 2
    count += 1   # One comparison made

    if arr[mid] == key:
        print(f"Found at index {mid} in {count} comparisons")
        return mid
    elif arr[mid] < key:
        return binary_search_recursive(arr, key, mid+1, high, count)
    else:
        return binary_search_recursive(arr, key, low, mid-1, count)

# Test
arr = [1, 3, 5, 7, 9, 11]
binary_search_recursive(arr, 7, 0, len(arr)-1)
# Output: Found at index 3 in 2 comparisons

binary_search_recursive(arr, 4, 0, len(arr)-1)
# Output: Not Found after 3 comparisons
```

Quick Revision — Important Formulas

Number System

- Trailing zeros in $n! = \lfloor n/5 \rfloor + \lfloor n/25 \rfloor + \lfloor n/125 \rfloor + \ldots$
- Sum of first n natural numbers $= n(n+1)/2$
- Sum of first n odd numbers $= n^2$
- Sum of first n even numbers $= n(n+1)$
- HCF $\times$ LCM = Product of two numbers

Profit, Loss & Percentages

- Profit% $= \frac{SP-CP}{CP} \times 100$
- If same SP with $x\%$ gain and $x\%$ loss: Net loss% $= x^2/100$
- Net discount for two successive discounts $a\%$ and $b\%$ $= a + b - ab/100$
- SP for desired profit% $= CP \times \frac{100+\text{Profit\%}}{100}$

Time, Speed, Distance & Work

- Speed = Distance/Time; Avg speed (equal dist) $= 2uv/(u+v)$
- Boat upstream $= B - S$; Downstream $= B + S$
- Work rate: If A completes in n days, rate $= 1/n$
- Pipes: Net fill rate $= \sum$(fill rates) $- \sum$(empty rates)

SI, CI, Mensuration

- SI $= PRT/100$; CI $= P(1 + r/100)^n - P$
- Diff (CI-SI) for 2 years $= P(r/100)^2$
- Circle area $= \pi r^2$; Cone CSA $= \pi rl$; Sphere vol $= \frac{4}{3}\pi r^3$
- Cylinder vol $= \pi r^2 h$; Cuboid vol $= l \times b \times h$

Permutation, Combination & Probability

- $P(n,r) = n!/(n-r)!$; $C(n,r) = n!/(r!(n-r)!)$
- Circular permutations $= (n-1)!$
- $P(\text{both same colour}) = \frac{\binom{a}{2}+\binom{b}{2}}{\binom{a+b}{2}}$
- $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

"Success is where preparation and opportunity meet."

Best of luck for TCS! — By Sanju

© 2026 — All Rights Reserved. Skill Bout & Sanju.